



Software Requirements Specification (SRS) Document for Fuel Management System (FMS)

August, 2025

Introduction

The **Fuel Management System (FMS)** is a software application designed to automate fuel refilling workflows, track consumption, enforce compliance, and generate detailed reports. This document outlines functional and non-functional requirements to ensure transparent, efficient, and fraud-resistant fuel management.

Here is general FMP steps :- Receive request for card refilling, officer will refill 50,000 - 100,000 birr to card via payment processing letter to finance at a time, then driver has to sign and receive card , every refilling time car km has to take photo to the system by driver with signed receipt back side, again drivers has to take photo of last km on dashboard and send to system for ending km with signed receipt back side, for next refilling officer will check vehicle mileage in k.m against standard and check available balance of E-card Record fuel balance, then the officer will Record daily fuel consumptions on data base Generate monthly report.

2. Functional Requirements

2.1 Fuel Card Setup & Replenishment

2.1.1 Description: The system shall allow administrators to pre-load funds (50,000–100,000 ETB) per vehicle's e-card and activate cards within 24 hours.

2.1.2 Inputs:

- Vehicle Plate Number
- E-card ID
- Fund Amount (ETB)
- Activation Timeframe (24hr SLA)
- Fuel consumed per vehicles
- Amount in Birr
- Beg Km

- End Km
- $D / KM = \text{end km} - \text{beg km}$
- $AVG \text{ KM/LITER} = D/km/\text{amount}$
- Standard
- Type of Vehicle
- All Cost center
- Branch
- District
- Reserve
- Pool vehicles
- Refill during maintenance process if the maintenance related to it
- Benzene
- Diesel

2.1.3 Outputs:

- E-card status (Active/Inactive).
 - Replenish fuel
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2.2 Fuel Refill Request

2.2.1 Description: Drivers shall submit refill requests with odometer readings.

2.2.2 Inputs:

- Vehicle Plate Number
- Current Odometer (km) (photo)
- All valid receipt
- Signed photo of all receipt
- Driver Cost Center
- Above standard

2.2.3 Outputs:

- Unique Request ID.
- Alert if mileage deviation exceeds -5% of standard km/L.

2.3 E-card Balance Validation

2.3.1 Description: The system shall validate e-card balance in real-time. Integrating with TOTAL card system.

2.3.2 Inputs:

- Request ID
- Plate number
- Driver name

2.3.3 Outputs:

- Approved/Declined status.
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2.4 Transaction Recording

2.4.1 Description: Capture actual dispensed fuel details.

2.4.2 Inputs:

- Liters used now (this information need integration with total cards)
- Balance in card now (this information need integration with total cards)

2.4.3 Outputs:

- Transaction of all drivers or individuals by plate
- Updated e-card balance.

2.5 Proof of Purchase Upload

2.5.1 Description: Drivers must upload fuel receipts and odometer photos.

2.5.2 Inputs:

- Receipt Image (JPG/PNG)
- Odometer Photo

- Digital Signature

2.5.3 Outputs:

- Documented audit trail.
- Alert for missing uploads.

2.6 Reporting

2.6.1 Daily Consumption Report

- **Inputs:** Odometer readings, fuel liters, cost.
- **Outputs:** Aggregated data per vehicle (km/L efficiency, cost).

Excel & PDF report containing the exact format shown below:

Monthly Fuel Consumption as of July 31/2025								
Vehicles Plate No	Fuel consumed per vehicles	Birr	Beg Km	End Km	D /K M		Stander	Type of Vehicle

2.6.2 Monthly Fuel Report

Summary of Fuel Expense from July 2024 upto June 2025

Types of Fuel	LITER	BIRR
TOTAL BENZENE	11,878.36	1,189,891.11
TOTAL DIESEL	207,119.42	20,542,732.01
TOTAL	218,997.78	21,732,623.12

Fuel expense by User Organ

No	User Organ	Fuel in Litres	Total in birr
1	Property Management	4,431.43	424,542.35
2	President Office	4,497.27	450,831.00
3	Motor Cycle	7,381.09	739,060.11
4	Construction Team	1,734.72	159,322.71
5	Security	2,520.29	193,667.56
6	Pool Service	198,432.98	19,765,199.39
	Total	218,997.78	21,732,623.12

5. Glossary

- **FRR:** Fuel Refill Request
- **D/KM:** Distance traveled (End Km – Beg Km)